

STROKE RISK PREDICTION

Healthcare Analytics and Machine Learning Project Report

Prepared by	Angeline Duarte
Technologies	Python, Pandas, Scikit-learn, Matplotlib, Tableau
Dataset	15,000 healthcare records
Report purpose	Portfolio presentation and management-level project summary

This report communicates the project's analytical workflow, findings, model performance, healthcare relevance, limitations, and recommended next steps.

Executive Summary

The Stroke Risk Prediction project evaluated a healthcare dataset containing 15,000 patient records to explore demographic, clinical, and lifestyle-related variables and assess whether machine learning could distinguish patients labeled as Stroke or No Stroke. The workflow included data-quality review, exploratory analysis, feature selection, an 80/20 train-test split, model training, comparative evaluation, and interpretation of results.

Three classification methods were tested: logistic regression, random forest, and a support vector machine. On the held-out test data, the models produced accuracy and weighted F1 scores near 0.50. These results indicate that the selected variables and model configuration did not provide reliable predictive discrimination in this version of the project. The project remains valuable as an end-to-end demonstration of data preparation, model comparison, visualization, evaluation, and responsible interpretation.

Management takeaway: The most important conclusion is not that the model is ready for clinical use, but that the analysis identified clear next steps: stronger preprocessing, feature engineering, validation, and data-quality assessment are required before the model could support decisions.

Project Snapshot

Records	Stroke Cases	Age Range	Missing Values
15,000	7,468 (49.8%)	18-90	0

1. Dataset Overview

The analysis used the primary data table in the supplied Excel workbook. After excluding an extra unnamed worksheet column, the working dataset contained 15,000 rows and 34 named fields.

- Target variable: Diagnosis, with 7,468 Stroke records and 7,532 No Stroke records.
- Patient age range: 18 to 90 years.
- Work types: Government Job, Never Worked, Private, Self-employed.
- Clinical variables included glucose, BMI, blood pressure, cholesterol, hypertension, heart disease, and stroke history.
- Lifestyle and contextual variables included smoking status, alcohol intake, physical activity, dietary habits, residence type, and stress level.
- Missing-value review: 0 missing values were identified in the primary working table.

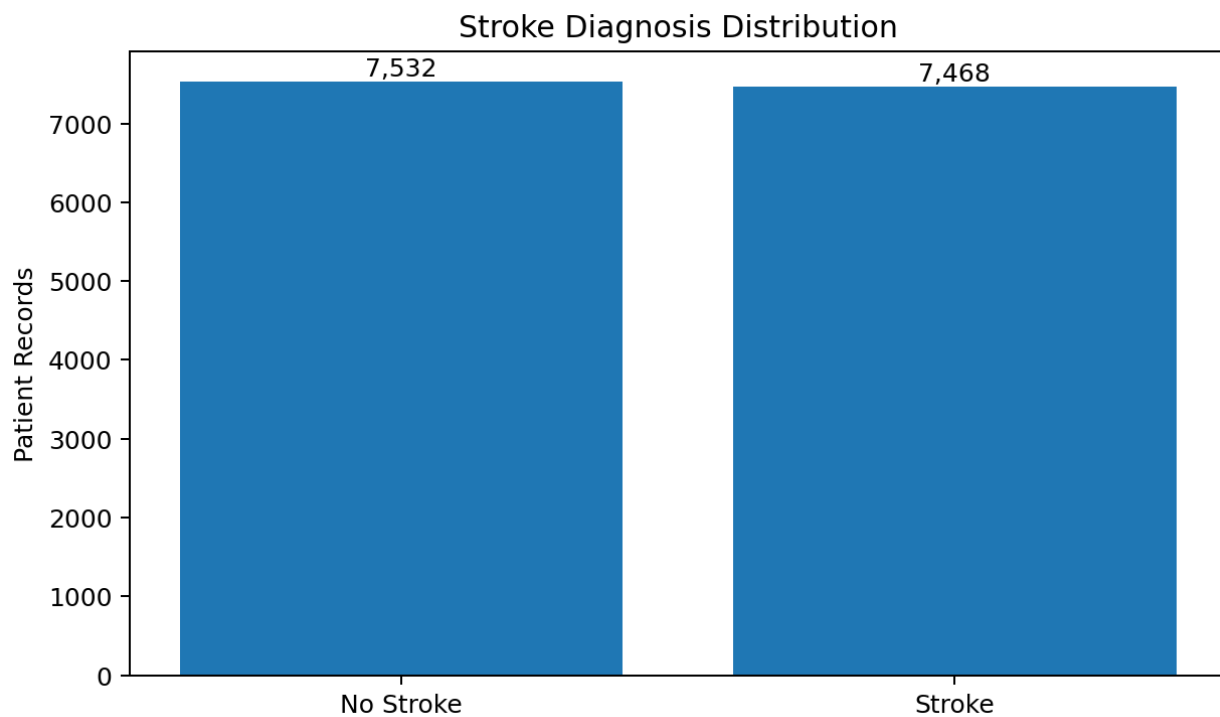


Figure 1. The target classes are nearly balanced in the working dataset.

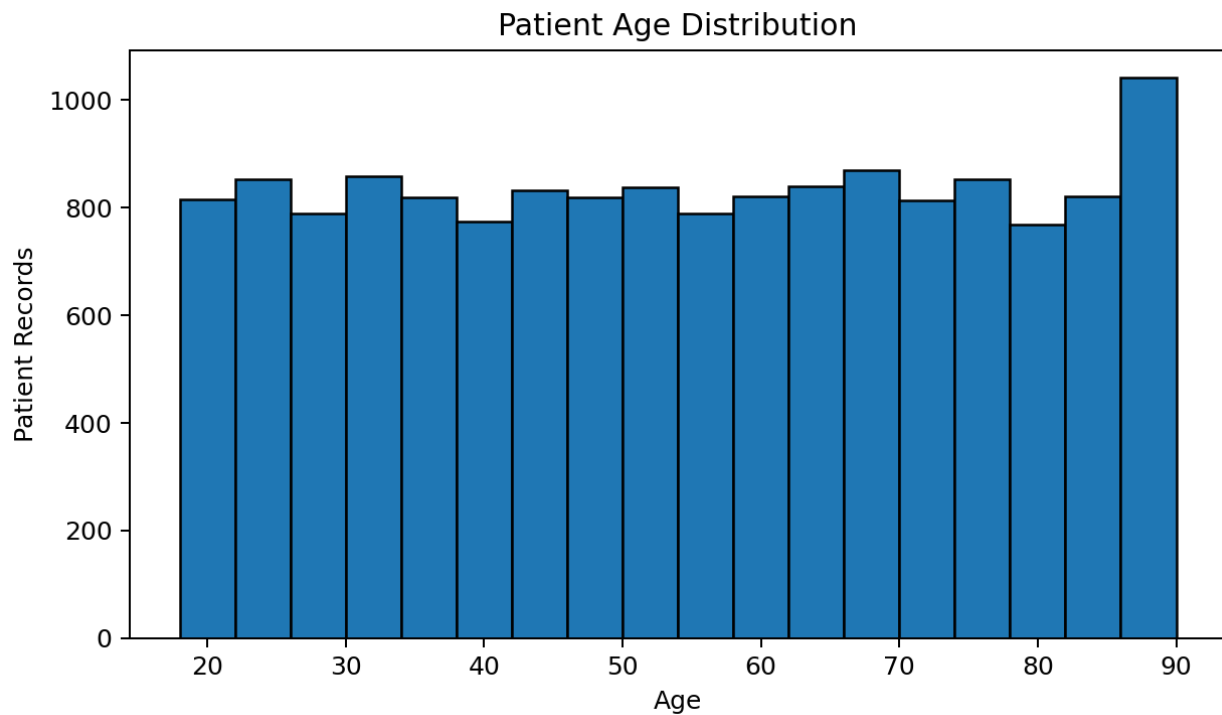


Figure 2. Ages span the full adult range, with broadly distributed observations.

Data Privacy and Governance

The workbook includes Patient ID and Patient Name fields. These fields were excluded from modeling because they do not provide appropriate predictive value and may create privacy or data-leakage concerns. Before publishing the dataset in a public GitHub repository, the names and identifying fields should be removed or the dataset should be replaced with a de-identified version whose public-sharing rights are confirmed.

Source Reconciliation Note

The original school Q&A referenced 15,001 records, while the primary analysis worksheet and Jupyter notebook contain 15,000 patient rows. This report uses the 15,000-row primary worksheet as the source of truth. The discrepancy appears to come from a pivot-table total rather than an additional patient record.

2. Methodology

Step	Phase	What was done
1	Data review	Inspected the workbook structure, column names, data types, categories, and missing values.
2	Feature preparation	Removed identifiers, the target, and selected categorical fields in accordance with the original notebook workflow.
3	Train-test split	Split the data into 80% training and 20% testing using random_state=42.
4	Model training	Trained logistic regression, random forest, and support vector machine classifiers.

5	Evaluation	Compared training accuracy, test accuracy, weighted F1, stroke precision, stroke recall, and confusion matrices.
6	Interpretation	Reviewed descriptive patterns, correlations, and random-forest feature importance while avoiding causal conclusions.

Analytical Scope

The report preserves the project's original modeling approach while adding a clearer and more professional interpretation of the results. It does not claim clinical validation, causal inference, or readiness for real-world patient care.

3. Exploratory Data Analysis

Exploratory analysis focused on the target distribution, age, work type, BMI, and correlations among selected numeric variables.

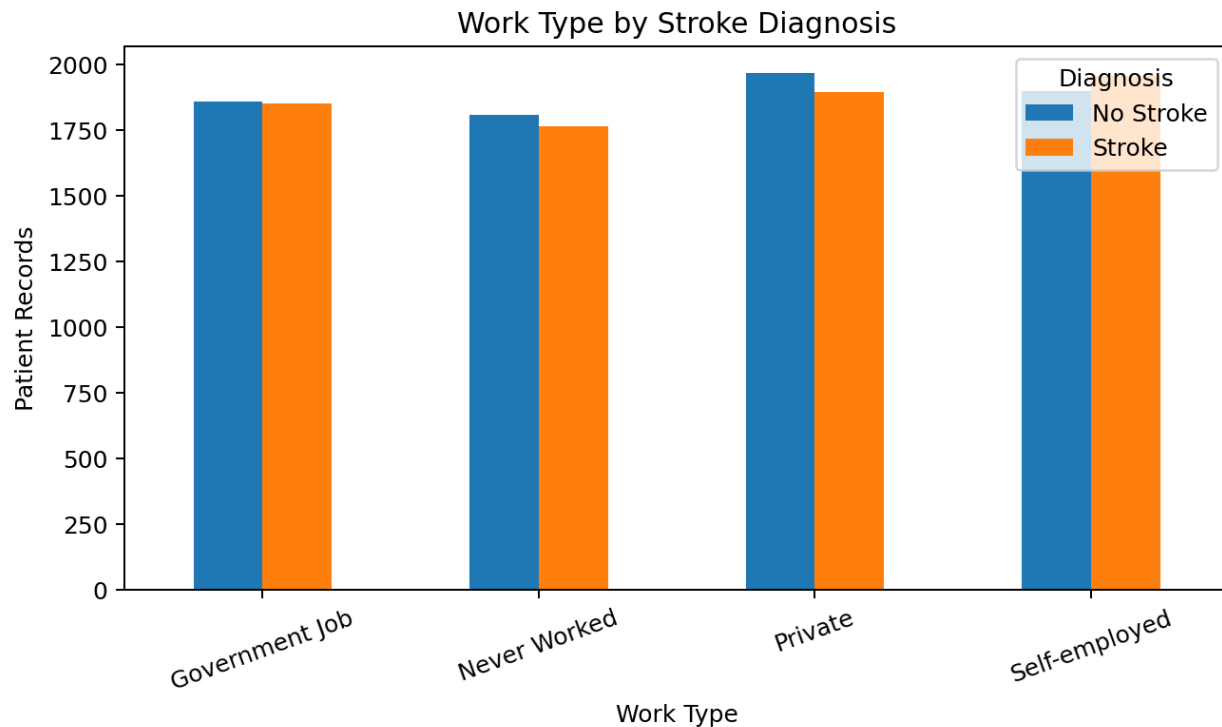


Figure 3. Work-type counts are similar across the two diagnosis classes.

Self-employed patients had the largest raw count of stroke records, but raw counts alone do not show that self-employment increases risk. The work-type categories contain different numbers of records, and no causal conclusion can be made without rate-based analysis and control for other factors.

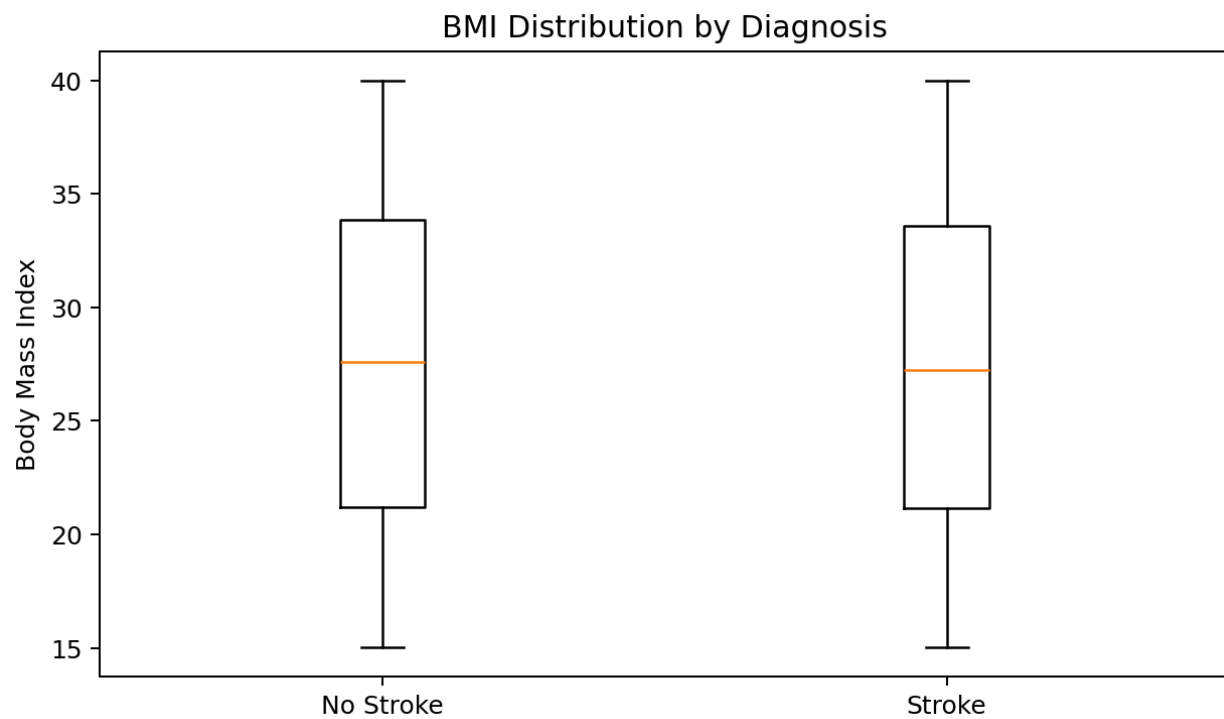


Figure 4. BMI distributions are highly similar across diagnosis groups.

The mean BMI was 27.40 for Stroke records and 27.55 for No Stroke records. This dataset therefore does not support the earlier claim that average BMI was substantially higher among stroke cases.

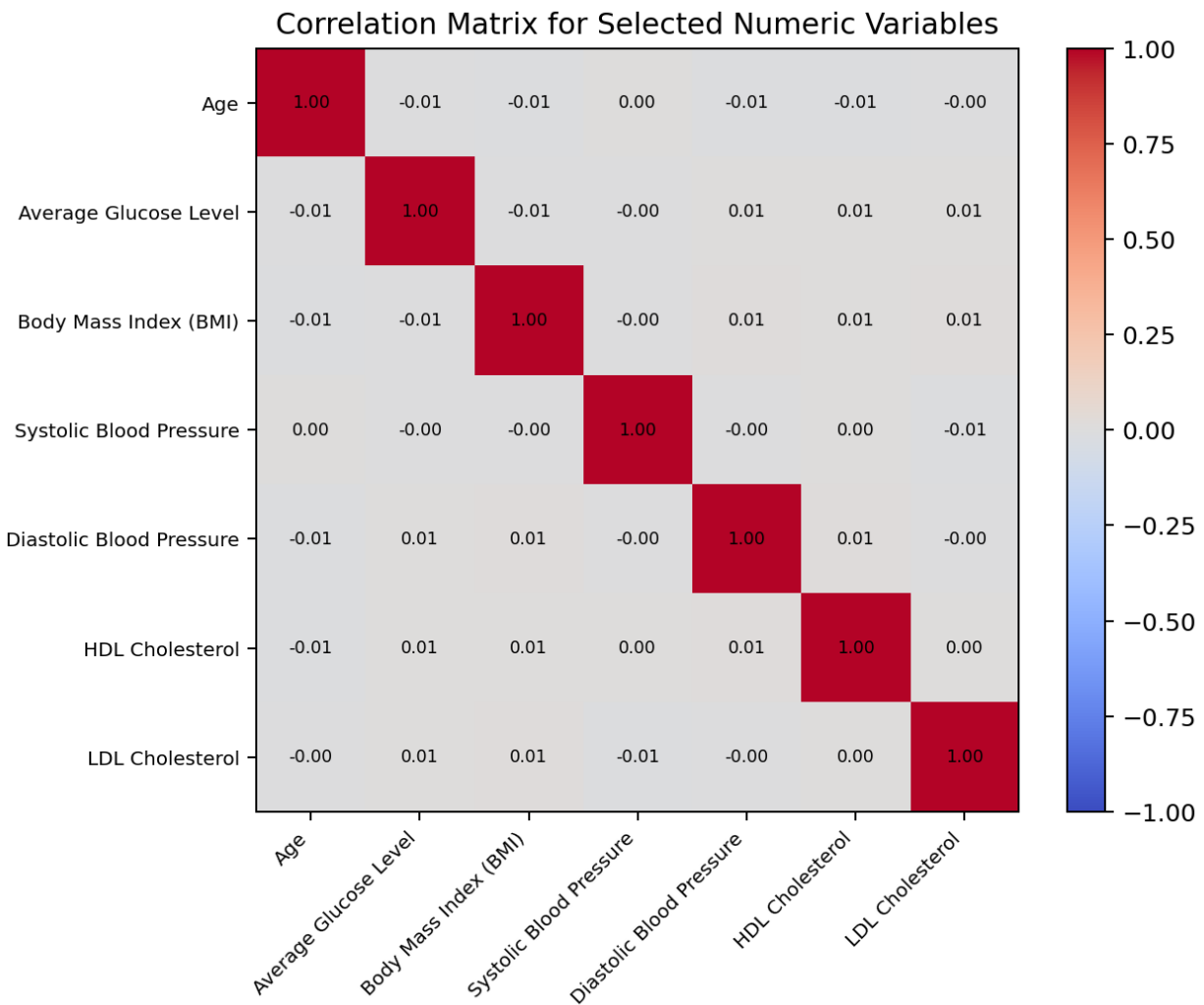


Figure 5. Selected numeric variables show mostly weak linear relationships.

The correlation matrix does not show uniformly strong negative relationships. Most coefficients are close to zero, which suggests limited linear association among the selected variables. Weak pairwise correlations do not rule out nonlinear or multivariable relationships.

4. Key Insights

- The diagnosis classes are nearly balanced, which reduces severe class-imbalance concerns but does not guarantee that the data contain usable predictive signal.
- The primary worksheet contains 15,000 records, not 15,001.
- No missing values were detected in the working table; however, completeness does not establish data validity or representativeness.
- Work-type differences are small relative to the category sizes and should not be interpreted causally.
- BMI averages and distributions are nearly identical between diagnosis groups in this dataset.
- Pairwise correlations among selected numeric fields are generally weak.
- The evaluated models perform close to chance on held-out data, indicating that the current pipeline is not suitable for clinical prediction.

Management takeaway: The project demonstrates analytical process and communication skills, while the weak predictive performance demonstrates the importance of reporting limitations honestly.

5. Machine Learning Results

The original notebook trained three classification models. The table below reports recalculated metrics from the supplied workbook using the same feature-selection and train-test approach.

Model	Train Acc.	Test Acc.	Weighted F1	Stroke Precision	Stroke Recall
Logistic Regression	0.516	0.488	0.487	0.488	0.446
Random Forest	1.000	0.503	0.502	0.504	0.459
Support Vector Machine	0.516	0.492	0.487	0.490	0.393

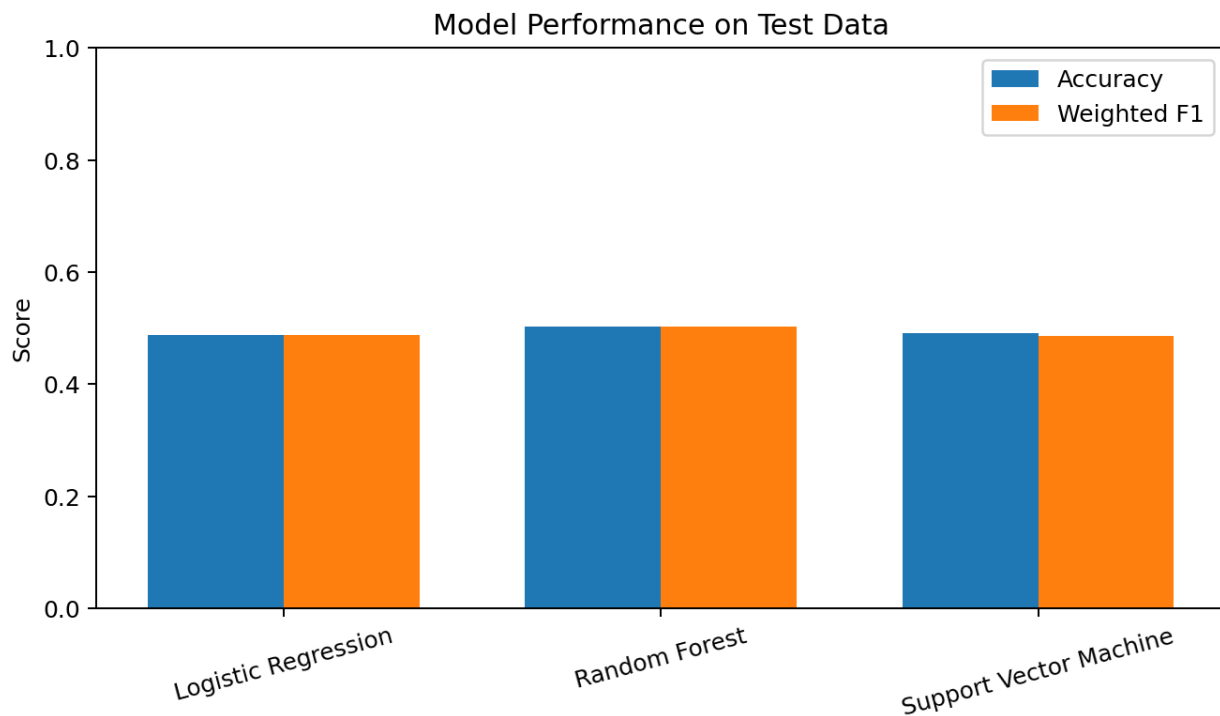


Figure 6. Test performance for all three models is near 0.50.

Interpretation

Random forest achieved perfect training accuracy but only about 0.50 test accuracy, a pattern consistent with overfitting. Logistic regression and SVM also performed near chance. In addition, the unscaled logistic regression generated a convergence warning during reproduction, suggesting that scaling and pipeline refinement should be part of the next iteration.

Confusion Matrices

Logistic Regression

Actual / Predicted	No Stroke	Stroke
No Stroke	795	703

Stroke	832	670
---------------	-----	-----

Random Forest

Actual / Predicted	No Stroke	Stroke
No Stroke	820	678
Stroke	813	689

Support Vector Machine

Actual / Predicted	No Stroke	Stroke
No Stroke	885	613
Stroke	912	590

6. Feature Importance and Model Signals

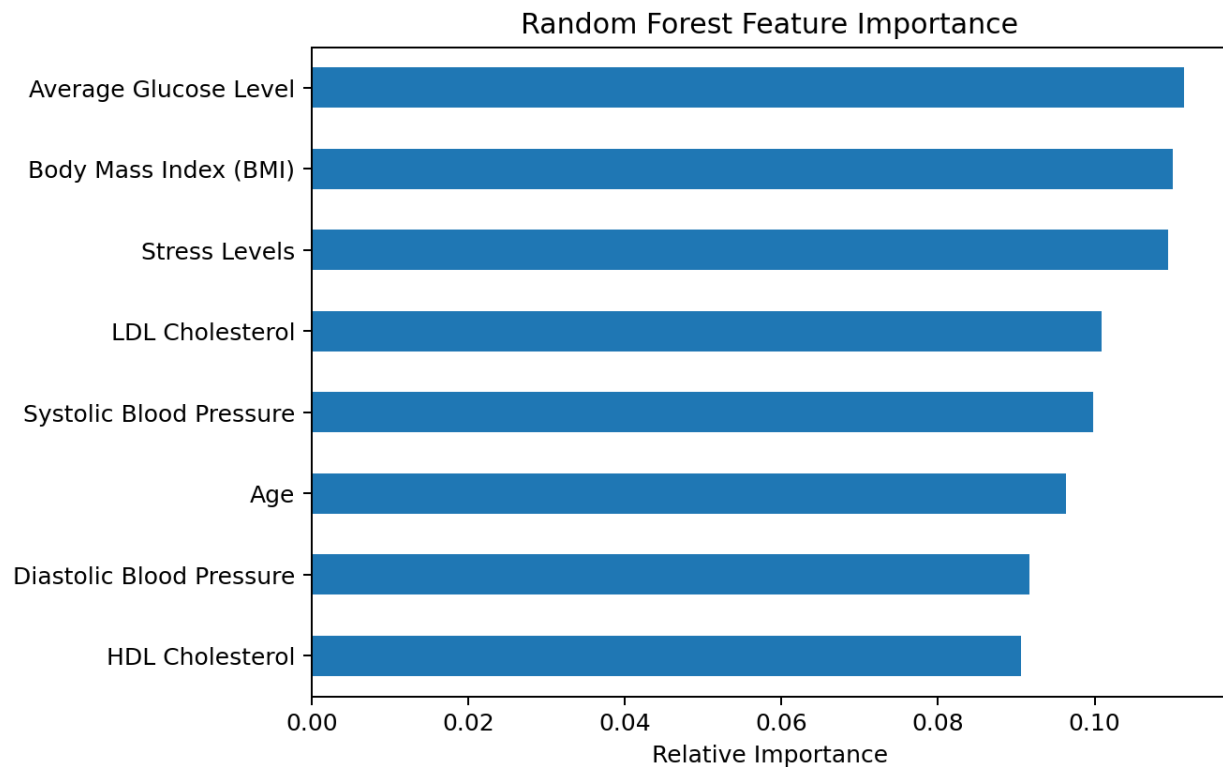


Figure 7. Relative feature importance from the reproduced random-forest model.

Average glucose level, BMI, stress level, cholesterol, blood pressure, and age received the largest relative importance values in the random-forest model. Feature importance indicates how the fitted model used variables; it does not prove clinical causation or that an individual feature is independently predictive.

Because the random-forest test performance was weak, these rankings should be treated as exploratory model signals rather than dependable clinical findings.

7. Potential Business Impact

A validated stroke-risk model could support population-health and healthcare-operations teams by helping prioritize outreach, identify groups for preventive education, and focus limited care-management resources. The current model is not validated for those uses, but the project demonstrates the workflow needed to move from raw healthcare data to evaluated analytical outputs.

- Risk-stratification support for care-management programs.
- Prioritization of preventive outreach and educational resources.
- Population-level trend monitoring.
- Analytical support for quality-improvement initiatives.
- Clear communication of model limitations to operational and clinical stakeholders.

8. Healthcare Relevance

Stroke prevention is a meaningful use case for healthcare analytics because many relevant factors are captured across clinical, demographic, and lifestyle data. Responsible use requires validated data, appropriate clinical oversight, transparent model behavior, privacy safeguards, and monitoring for bias and performance drift.

9. Limitations

- The dataset's provenance and representativeness were not documented in the supplied files.
- The nearly 50/50 diagnosis distribution may not reflect real-world stroke prevalence and could indicate synthetic or rebalanced data.
- The primary model workflow removed several categorical variables rather than encoding and evaluating them.
- Numeric features were not standardized before logistic regression and SVM.
- The analysis used one train-test split rather than cross-validation.
- Random forest substantially overfit the training data.
- No hyperparameter tuning, calibration analysis, ROC-AUC assessment, subgroup analysis, or external validation was included.
- The project does not establish causation and is not appropriate for patient-level decision-making

10. Future Improvements

- Create a reproducible preprocessing pipeline with scaling and categorical encoding.
- Remove or de-identify patient names and IDs before public publication.
- Use stratified cross-validation and hyperparameter tuning.
- Evaluate ROC-AUC, PR-AUC, sensitivity, specificity, calibration, and confidence intervals.
- Test class-weighting and threshold selection based on the intended healthcare use case.
- Compare logistic regression with tuned random forest, gradient boosting, and XGBoost models.
- Use SHAP or permutation importance for more transparent model interpretation.
- Assess performance across demographic and clinical subgroups to identify potential bias.
- Validate the approach on an independent, documented healthcare dataset.

- Create an interactive dashboard that clearly separates descriptive findings from model predictions.

11. Project Deliverables

Deliverable	Purpose
Jupyter notebook	Documents data preparation, model training, evaluation, and visualizations.
Excel dataset	Contains the source records and supporting worksheet outputs.
Project report	Communicates the workflow, evidence, interpretation, limitations, and next steps.
GitHub README	Provides a concise recruiter-facing overview and repository navigation.
Portfolio page	Presents the project visually and links to the technical materials.

12. Conclusion

The Stroke Risk Prediction project demonstrates a complete introductory healthcare analytics workflow: data review, exploratory visualization, feature preparation, classification modeling, comparative evaluation, and communication of results. The strongest professional lesson from the project is the importance of honest model assessment. Although the models did not achieve reliable predictive performance, the analysis identified specific technical and governance improvements required for a stronger future version.

Presented in this form, the project shows both technical capability and judgment: the ability to build models, interpret results cautiously, communicate findings to nontechnical stakeholders, and distinguish a learning prototype from a production-ready healthcare solution.